CALIBER 2026 | CASE 1: MANUFACTURING KNOWLEDGE HUB

# Business Impact and Financial Modelling Report

## Asset Knowledge Platform


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**IDR 3.8 billion**
annual value at full rollout, 3 units

**IDR 3.3 billion**
net present value before tax, 3 units

**34 months**
payback, 3 units

**82%**
probability of a positive NPV


Version 1.0 | 3 October 2026 | Companion to Business_Impact_Financial_Model.xlsx. Base scenario. Sample data provided for CALIBER purposes only.

# Contents

1 Executive Summary ..... 2
2 Context, Scope and Methodology ..... 3
3 Market Analysis ..... 4
4 Business Impact Model ..... 6
5 Investment and Financial Returns ..... 8
6 Uncertainty and Risk ..... 10
7 Benefits Realisation and Conclusions ..... 12
8 Appendix: Key Assumptions and Reconciliation ..... 13

# 1 Executive Summary

> A programme covering **three units** returns **IDR 3.8 billion a year** at full rollout and a net present value of **IDR 3.3 billion** before tax. The outcome is positive in **82%** of 5,000 simulated scenarios.

This report sets out the financial case for the Asset Knowledge Platform (AKP), the Manufacturing Knowledge Hub proposed in response to the CALIBER 2026 Case Book. It replaces the single business impact slide of the proposal with a complete financial model. Every figure now traces to a formula in the accompanying workbook, **Business_Impact_Financial_Model.xlsx** (23 sheets, about 2,187 live formulas and 28 integrity checks), to the 211 work orders supplied with the case, or to a cited public source.

The Base scenario recognises four benefit levers that rest on evidence: time spent searching for and checking sources, repair time, repeat failures and a faster start for new staff. Start-up delay, incident cost and document rework are reported separately as upside and are excluded from every headline figure. All values are stated in Indonesian rupiah (IDR) at year 0 prices and before tax, unless stated otherwise.

| Base scenario | 1 unit (pilot) | 3 units | 6 units | 12 units |
|---|---|---|---|---|
| Annual value at full rollout (IDR million) | 1,275 | 3,825 | 7,458 | 14,533 |
| One-off cost over the programme (IDR million) | 1,766 | 2,708 | 4,148 | 7,067 |
| Net present value before tax (IDR million) | -205 | 3,320 | 7,446 | 14,131 |
| Net present value after tax (IDR million) | -252 | 2,442 | 5,560 | 10,550 |
| Internal rate of return before tax | 8% | 44% | 62% | 77% |
| Payback period (months) | 50 | 34 | 32 | 34 |
| Probability of a positive NPV (Monte Carlo) | 27% | 82% | 93% | 97% |

## 1.1 Principal findings

- **The pilot unit does not repay on its own.** One unit returns IDR 1.3 billion a year against a one-off cost of IDR 1.8 billion, giving an NPV of -IDR 0.2 billion. The pilot buys evidence and a reusable platform.
- **Value is created by scale.** The shared cost of IDR 1.3 billion is paid once. The NPV turns positive from two units and reaches IDR 14.1 billion at twelve units. Chandra Asri operates about 21 unit equivalents, so twelve units represent 58% of its scale.
- **The downside is bounded.** Across 5,000 simulated outcomes the three-unit NPV has a 5th percentile of -IDR 1.4 billion and a 95th percentile of IDR 7.6 billion. The dominant drivers are the share of breakdown hours that reduce output, the margin per tonne and the pay level used to value time.
- **The market is a value pool, not a revenue pool.** Indonesian chemical and petrochemical manufacturing holds about 480 process units, with a customer value pool of IDR 611.9 billion a year. A vendor could capture only about IDR 122,387 million (6.8 million USD), so the investment case rests on value created inside the Company.
- **Four earlier assumptions have been corrected.** The cost anchor, the value of a downtime hour, the share of freed time that is productive and the share of the gain credited to the platform are now explicit and sourced (Appendix, Section 8).

> **NOTE**
>
> **Recommendation.** Approve the pilot at a one-off cost of IDR 1,766 million as the first stage of a programme of at least three units, governed by the three stage gates in Section 7.

# 2 Context, Scope and Methodology

The CALIBER 2026 Case Book asks for a Manufacturing Knowledge Hub that connects scattered plant knowledge, gives engineers trusted and cited answers, and supports reliability decisions. The business impact section of the proposal must show the expected benefits, the key performance indicators and the measurable value, with supporting evidence. This report addresses the business side of the three key questions of the Case Book.

| Key question | Business answer in this report |
|---|---|
| 1. How can the Company build a structured Industrial Data Ops foundation? | A shared one-off cost of **IDR 1,322 million** covers four connectors, security and governance. Section 5 prices it and shows that it is repaid from two units onwards. |
| 2. How can AI help engineers find trusted information faster? | Time spent searching and checking falls by **50%**, from 40 to 20 minutes a day. Lever 1 values this at **IDR 340 million a year** for one unit. |
| 3. How can the hub support reliability and continuous improvement? | Failure memory shortens repairs and prevents repeat failures. Levers 2 and 3 account for **70%** of the value. |

## 2.1 Methodology

1. **Baseline.** Breakdowns, downtime, repeat failures and repair cost are re-derived from the 211 work orders, using formulas applied to the raw data.
2. **Value.** Time saved, repair hours saved and failures avoided are converted into IDR using an explicit loaded labour cost and a bottom-up value of one downtime hour.
3. **Cost.** One-off, per-unit and running costs are built line by line, with a contingency.
4. **Return.** The rollout is phased by cohort, an adoption ramp is applied, and cash flows are discounted at the weighted average cost of capital of **11.8%**. NPV, IRR, payback and benefit-cost ratio are reported before and after tax.
5. **Stress.** One-at-a-time sensitivities, break-even values, a Monte Carlo simulation and a risk register test the result.
6. **Verification.** The workbook is recalculated, checked by 28 integrity tests and compared with an independent Python replica.

Every input carries an evidence tag: **Case data**, **Market source**, **Estimate** or **Assumption**. The Inputs sheet lists the pessimistic, base and optimistic value of each driver, and a single switch changes the whole workbook between the three scenarios. The twelve most influential inputs are listed in the Appendix.

# 3 Market Analysis

> The Company's own economics, Indonesian demand and the technology market point in the same direction: availability and trusted knowledge become more valuable each year.

## 3.1 The Company and the Indonesian market

| Indicator | Value | Source |
|---|---|---|
| Polyethylene capacity at Cilegon | 736 kt a year (336 kt existing, 400 kt added in 2020) | Chemical Engineering; Chandra Asri |
| Ethylene capacity | 860 kt a year; the only producer in Indonesia | Chandra Asri investor update, 2018 |
| Employees, 2024 | 3,746 | Sustainability report 2024 |
| Revenue and result, FY2024 | USD 1.79 billion, down 17.4%; net loss of USD 69 million | IDNFinancials |
| Polyethylene and polypropylene demand | 3.6 Mt a year, rising by about 50% to 5.3 Mt by 2027; half is imported | AHK Indonesia |
| Domestic polyethylene capacity | 1.2 Mt a year at about 80% utilisation; import quotas since March 2024 | AHK Indonesia; Argus |
| Chemical industry, 2024 | Exports of USD 17.4 billion; investment of IDR 65.8 trillion | Ministry of Industry |
| Southeast Asia LLDPE price, 2026 | USD 920 to 1,500 per tonne CFR | Polymer Update |

Three conclusions follow. **First**, Chandra Asri reported a net loss in FY2024 in a volatile margin environment, so a digital programme must repay quickly and be released in stages. **Second**, Indonesia imports about half of its polyethylene and is raising import barriers, so every point of domestic utilisation matters: one percentage point on 1.2 Mt of capacity equals **12 kt**, or about **USD 2.6 million** of margin at USD 220 per tonne. **Third**, plant knowledge is at risk: surveys report that about **40%** of chemical workforces are aged 45 or over and that most of their knowledge is tacit.

## 3.2 Technology and services market

- Indonesia's AI market is estimated at **EUR 2.2 billion** in 2024 and **EUR 10.2 billion** in 2030, a compound growth rate of 28.9% a year (Statista).
- Global knowledge management software was **USD 20.2 billion** in 2024 and grows at 13.6% a year (Grand View Research). Asset performance management was about USD 24.8 billion.
- Predictive maintenance in Asia Pacific is **USD 3.8 billion** in 2026 and reaches USD 7.5 billion in 2031, growing at 14.4% a year.
- Knowledge workers spend **1.8 hours** a day searching for information (McKinsey) and up to 2.5 hours (IDC). The model assumes only **40 minutes** a day for plant staff.

## 3.3 Sizing the opportunity

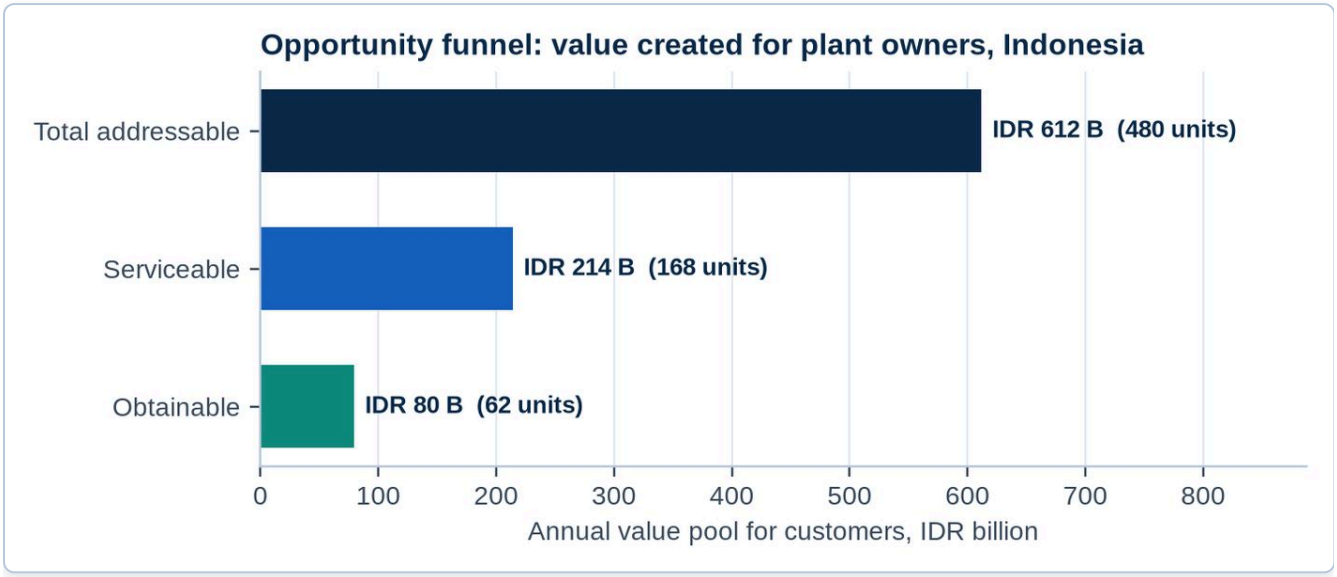



**Figure 1. Opportunity funnel for Indonesian chemical and petrochemical manufacturing. Source: Market sheet of the workbook.**

The bottom-up sizing assumes 80 manufacturing sites with 6 process units each, which gives **480 units**. A unit is a process area comparable with the sample LLDPE unit, with about 45 plant staff. Of these, 35% sit at sites with an existing document or maintenance system and a digital programme (**168 units**), and **62 units** could be reached within five years: 12 internal units and about 50 external ones. At the value per unit of IDR 1,275 million a year, the value pool falls from **IDR 611.9 billion** to **IDR 214.2 billion** and **IDR 79.6 billion**. If a provider priced at 20% of the value created, the Indonesian revenue pool would be **IDR 122,387 million** (6.8 million USD), or about 27 million USD with a Southeast Asia multiple of four. A top-down estimate from the national AI market gives a serviceable pool within a factor of 8 of this figure.

> **NOTE**
> The external revenue pool is small. The platform should therefore be funded for the value it creates inside the Company. Commercial extension is an option and not the basis of the investment case.

## 3.4 Alternatives and positioning

**Alternatives and positioning**

*Team assessment on a 1 to 5 scale (5 is best). Scores are judgements to be tested with the Company. Only the list price of the generic assistant is a published figure.*

| Criterion | Weight | This platform | Generic workplace assistant | Enterprise document search | EAM or CMMS vendor module | In-house custom build |
|---|---|---|---|---|---|---|
| Cited answers with approval status and confidence | 20% | 5 | 3 | 2 | 3 | 3 |
| Tacit knowledge capture and approval workflow | 15% | 5 | 2 | 1 | 2 | 3 |
| Failure memory linked to work orders and lessons | 15% | 5 | 1 | 1 | 4 | 3 |
| Plant context: P&ID, interlock, drawing, 3D | 15% | 5 | 1 | 2 | 3 | 3 |
| Control of data and deployment | 10% | 4 | 2 | 4 | 4 | 5 |
| Time to value | 10% | 4 | 4 | 3 | 2 | 1 |
| Five-year cost (5 is lowest) | 10% | 4 | 3 | 4 | 2 | 2 |
| Independence from one vendor | 5% | 4 | 2 | 3 | 2 | 4 |
| **Weighted score** | **100%** | **4.65** | **2.20** | **2.25** | **2.85** | **2.95** |
| Rank | | 1 | 5 | 4 | 3 | 2 |

**Figure 2. Weighted comparison of alternatives (Competitors sheet). Scores are the team's assessment on a scale of 1 to 5.**

A generic workplace assistant is the closest low-cost alternative. At the published list price of USD 30 per user a month, 135 users cost about **IDR 4,350 million** over five years, before the base Microsoft 365 licence. That is lower than the **IDR 8,900 million** five-year cost of the platform. However, a generic assistant does not hold approval status, interlock logic, work order history or tacit knowledge as structured and auditable facts, and these features create most of the value in Levers 2 and 3.

# 4 Business Impact Model

> The model starts from what the work orders show and values four effects. Each step is a formula that can be inspected.

## 4.1 Baseline from the case data

**Case metrics**

*Every figure is a formula on Case_Data. These are the only values taken directly from the case data in the business impact model.*

| Metric | Value | Definition |
|---|---|---|
| History start (first report date) | 2024-06-04 | Earliest work order report date. |
| History end (platform cut-off) | 2025-12-10 | Cut-off used by the platform. |
| Days covered | 554 | End minus start. |
| Years covered | 1.52 | Days divided by 365. |
| Work orders | 211 | All work orders in the file. |
| Breakdowns | 31 | Breakdown flag equal to Yes. |
| Breakdown downtime (hours) | 434 | Sum of downtime hours on breakdown work orders. |
| Mean time to repair (hours) | 14.0 | Downtime hours per breakdown. |
| Breakdowns a year | 20.4 | Annualised. |
| Breakdown downtime a year (hours) | 286 | Breakdowns a year times MTTR. |
| Corrective or breakdown jobs | 63 | Work type Corrective, or breakdown flag Yes. |
| Corrective or breakdown jobs a year | 41.5 | Annualised. |
| Breakdown share of those jobs | 49.2% | Breakdowns divided by corrective or breakdown jobs. |
| Jobs that repeat within 120 days | 13 | Same equipment tag and same failure mode as an earlier job in the previous 120 days. |
| Repeat rate | 20.6% | Repeating jobs divided by corrective or breakdown jobs. |
| Average direct cost of a corrective or breakdown job (IDR M) | 6.99 | Labour plus material cost. |

**Figure 3. Case_Metrics sheet. Every figure is a formula applied to the 211 work orders.**

Between June 2024 and December 2025 the plant recorded **31 breakdowns**, which cost **434 hours** of downtime. The mean time to repair is **14.0 hours** and breakdowns occur at **20.4 a year**, equal to 286 downtime hours a year. Of 63 corrective or breakdown jobs, **13 (21%)** repeat the same failure mode on the same equipment within 120 days. A direct repair job costs **IDR 7.0 million** on average. In addition, 66 of the 98 controlled documents carry no approval status.

## 4.2 Value of one hour of downtime

The earlier slide used a flat IDR 30 million per hour. The revised value starts from a unit capacity of 336 kt a year, equal to 42 tonnes an hour, and a lost margin of USD 220 per tonne. A full stoppage then costs **IDR 165 million** an hour. Many equipment sets have installed spares or only reduce performance when they fail, so only **20%** of breakdown hours are assumed to reduce output. This gives **IDR 33.1 million** per hour. It is the most influential assumption in the model and is tested in Section 6.

## 4.3 Four value levers

**Value levers per unit**

*Annual value of one unit at 100% adoption, in IDR M at year 0 prices. Adoption, rollout and escalation are applied on the cash flow sheets.*

| Calculation step | Value | Unit | Evidence | Basis |
|---|---|---|---|---|
| **LEVER 1: SEARCH AND CHECKING TIME** | | | | |
| Users per unit | 45 | people | Assumption | Labour sheet. |
| Minutes saved per user a day | 20.0 | minutes | Assumption | Baseline minutes times reduction. |
| Hours saved per user a year | 76.7 | hours | Derived | Minutes saved times working days. |
| Average loaded cost (IDR an hour) | 193,500 | IDR | Estimate | Labour sheet. |
| Share of freed time that becomes productive | 60% | % | Assumption | Conversion factor. |
| **Lever 1 value (IDR M a year)** | **400.5** | IDR M | Assumption | Users times hours times cost times conversion. |
| **LEVER 2: REPAIR TIME** | | | | |
| Breakdowns a year | 20.4 | events | Case data | Case_Metrics. |
| Repeat breakdowns avoided a year | 1.26 | events | Derived | Breakdowns times repeat rate times reduction. |
| Remaining breakdowns a year | 19.16 | events | Derived | Avoided repeats are removed first so that lever 3 is not counted twice. |
| Mean time to repair (hours) | 14.0 | hours | Case data | Case_Metrics. |
| Reduction in repair time | 10.0% | % | Assumption | Input. |
| Share credited to the platform | 70% | % | Assumption | Input. |
| Value of one breakdown hour (IDR M) | 33.1 | IDR M | Estimate | Downtime_Value. |
| **Lever 2 value (IDR M a year)** | **621.1** | IDR M | Assumption | Remaining breakdowns times MTTR times reduction times credit times hour value. |
| **LEVER 3: REPEAT FAILURES** | | | | |
| Repeat rate of corrective jobs | 20.6% | % | Case data | Case_Metrics. |
| Corrective jobs a year | 41.5 | jobs | Case data | Case_Metrics. |
| Repeat jobs avoided a year | 2.57 | jobs | Derived | Jobs times repeat rate times reduction. |
| Of which breakdowns | 1.26 | events | Derived | Same breakdowns as in lever 2. |
| Downtime avoided (IDR M) | 585.5 | IDR M | Derived | Avoided breakdowns times MTTR times hour value. |
| Direct repair cost avoided (IDR M) | 18.0 | IDR M | Case data | Avoided jobs times average cost. |
| **Lever 3 value (IDR M a year)** | **422.4** | IDR M | Assumption | Sum credited to the platform. |
| **LEVER 4: FASTER START FOR NEW AND ROTATED STAFF** | | | | |
| New joiners and role changes a year | 6.0 | people | Assumption | Input. |
| Days saved per joiner | 10.0 | days | Assumption | Input. |
| Loaded cost of a working day (IDR) | 1,548,000 | IDR | Estimate | Hourly cost times hours a day. |
| **Lever 4 value (IDR M a year)** | **55.7** | IDR M | Assumption | Joiners times days times cost times conversion. |
| **HEADLINE TOTAL AT 100% ADOPTION (IDR M a year)** | **1,499.8** | IDR M | Mixed | Levers 1 to 4. |
| **Total at steady-state adoption (IDR M a year)** | **1,274.9** | IDR M | Assumption | Multiplied by steady-state adoption. |
| **UPSIDE LEVERS (NOT IN THE HEADLINE)** | | | | |
| Upside A: start-up delay avoided (IDR M a year) | 231.6 | IDR M | Assumption | Events times hours times full-stop hour value times credit. |
| Upside B: incident cost avoided (IDR M a year) | 150.0 | IDR M | Assumption | Frequency times cost times reduction. |
| Upside C: document rework avoided (IDR M a year) | 28.4 | IDR M | Assumption | Documents times hours times reduction times cost. |
| **Total upside at 100% adoption (IDR M a year)** | **410.0** | IDR M | Assumption | Not added to the headline. |

**Figure 4. Levers sheet. Annual value of one unit at full adoption, in IDR million at year 0 prices.**

| Lever | How it is valued | Evidence | IDR million a year |
|---|---|---|---|
| 1. Search and checking time | 45 users x 20 minutes saved x 230 days x loaded cost x 60% productive | Assumption; cited context | 340 |
| 2. Repair time | Remaining breakdowns x 14.0 hours x 10% x 70% credited x IDR 33 million per hour | Case data; estimate | 528 |
| 3. Repeat failures | Avoided repeat jobs (2.6 a year) x repair cost and downtime, 70% credited | Case data | 359 |
| 4. Faster start for new staff | 6 joiners x 10 days x loaded day cost x 60% productive | Assumption | 47 |
| **Total at 85% adoption** | | | **1,275** |

Avoided repeat breakdowns are removed before the repair time lever is calculated, so that no hour is counted twice. The upside levers are start-up delay avoided (**IDR 197 million**), incident cost avoided (**IDR 128 million**) and document rework (**IDR 24 million**) per unit and year. They rest on assumptions only and are excluded from every headline figure.

# 5 Investment and Financial Returns

## 5.1 One-off and running cost

**Cost model**

*One-off cost shared by all units, one-off cost for each unit, and running cost. IDR M at year 0 prices.*

| Item | Value (IDR M) | Unit | Basis |
|---|---|---|---|
| **SHARED ONE-OFF COST** | | | |
| Platform configuration and hardening | 440.0 | IDR M | Person-months times blended rate. |
| Source system connectors | 400.0 | IDR M | Connectors times cost per connector. |
| Security review and penetration test | 150.0 | IDR M | Input. |
| Governance setup and change design | 100.0 | IDR M | Input. |
| Training material and demo content | 60.0 | IDR M | Input. |
| **Subtotal before contingency** | **1,150.0** | IDR M | |
| Contingency | 172.5 | IDR M | Percentage of subtotal. |
| **Shared one-off cost** | **1,322.5** | IDR M | Incurred in year 0. |
| **ONE-OFF COST FOR EACH UNIT** | | | |
| Document ingestion, tagging and cleansing | 147.0 | IDR M | Person-days times cost. |
| Knowledge capture sessions | 144.0 | IDR M | Sets times sessions times cost. |
| User training time | 34.8 | IDR M | Users times hours times average loaded cost. |
| Validation and acceptance testing | 60.0 | IDR M | Input. |
| **Subtotal before contingency** | **385.8** | IDR M | |
| Contingency | 57.9 | IDR M | Percentage of subtotal. |
| **One-off cost for each unit** | **443.7** | IDR M | Incurred the year before the unit goes live. |
| **AI SERVICE USAGE COST** | | | |
| Questions a year per unit | 82,800 | queries | Users times questions a day times days. |
| Input tokens a year (millions) | 248.4 | M tokens | Questions times tokens per question. |
| Output tokens a year (millions) | 49.7 | M tokens | Questions times tokens per answer. |
| Service cost a year (USD) | 596 | USD | Tokens times list price times buffer. |
| **Service cost a year (IDR M)** | **10.7** | IDR M | Converted at the exchange rate. |
| **RUNNING COST A YEAR** | | | |
| Cloud hosting and storage | 80.0 | IDR M | Input. |
| AI platform agreement and data residency | 90.0 | IDR M | Input. |
| Knowledge manager | 240.0 | IDR M | FTE times monthly loaded cost times 12. |
| Platform engineer | 144.0 | IDR M | FTE times monthly loaded cost times 12. |
| Annual security audit | 60.0 | IDR M | Input. |
| Monitoring, licences and tools | 40.0 | IDR M | Input. |
| AI service usage | 10.7 | IDR M | From the usage calculation above. |
| **Subtotal before contingency** | **664.7** | IDR M | |
| Contingency | 66.5 | IDR M | Percentage of subtotal. |
| **Running cost for the first unit** | **731.1** | IDR M | Year 0 prices. |

**Figure 5. Costs sheet. Shared one-off cost, per-unit one-off cost and running cost, in IDR million at year 0 prices.**

The shared one-off cost is **IDR 1,322 million**, including a 15% contingency. It covers platform configuration (8 person-months), four connectors (EDMS, AIMS, CMMS and the historian), a security review, governance and training material. Each unit then requires **IDR 444 million** for document ingestion, knowledge capture sessions, user training and acceptance testing. Running cost is **IDR 731 million a year** for the first unit and **IDR 256 million** for each additional unit.

The AI service itself costs only about IDR 11 million a year for one unit at list prices and is not a cost driver. People, data preparation and integration are. The pilot one-off cost of **IDR 1,766 million** is 26% above the earlier estimate of IDR 1,400 million, and the running cost is 98% above the earlier IDR 370 million. The earlier figures had no line-by-line support.

## 5.2 Returns by programme size

**Summary of results**

*Live links to the cash flow sheets. Change the scenario on Inputs!D4 to see the other cases.*

Acti

| Metric (IDR M unless stated) | 1 unit (pilot) | 3 units | 6 units | 12 units |
|---|---|---|---|---|
| **Steady-state annual value (year 0 prices)** | **1,275** | **3,825** | **7,458** | **14,533** |
| Total one-off cost | 1,766 | 2,708 | 4,148 | 7,067 |
| Running cost in year 5 (year 0 prices) | 731 | 1,243 | 2,011 | 3,546 |
| Total benefits over 5 years | 6,444 | 15,212 | 26,193 | 45,280 |
| Total costs over 5 years | 5,885 | 8,900 | 13,026 | 20,750 |
| Net benefit over 5 years, before tax | 559 | 6,312 | 13,167 | 24,530 |
| **NPV before tax** | **(205)** | **3,320** | **7,446** | **14,131** |
| NPV after tax | (252) | 2,442 | 5,560 | 10,550 |
| IRR before tax | 8% | 44% | 62% | 77% |
| IRR after tax | 7% | 38% | 54% | 66% |
| **Payback (months)** | **49.9** | **33.9** | **32.2** | **33.7** |

**Figure 6. Summary sheet. Results by programme size, Base scenario, IDR million unless stated.**

A three-unit programme requires **IDR 2,708 million** of one-off spend in total, because units two and three are onboarded in years 1 and 2. It reaches **IDR 3,825 million** of annual value at full rollout and a net present value of **IDR 3,320 million** before tax (**IDR 2,442 million** after tax). The internal rate of return is **44%** before tax and the benefit-cost ratio is **1.48**. Simple payback is **34 months** and discounted payback is 38 months, because adoption builds over two years and the later units start later.

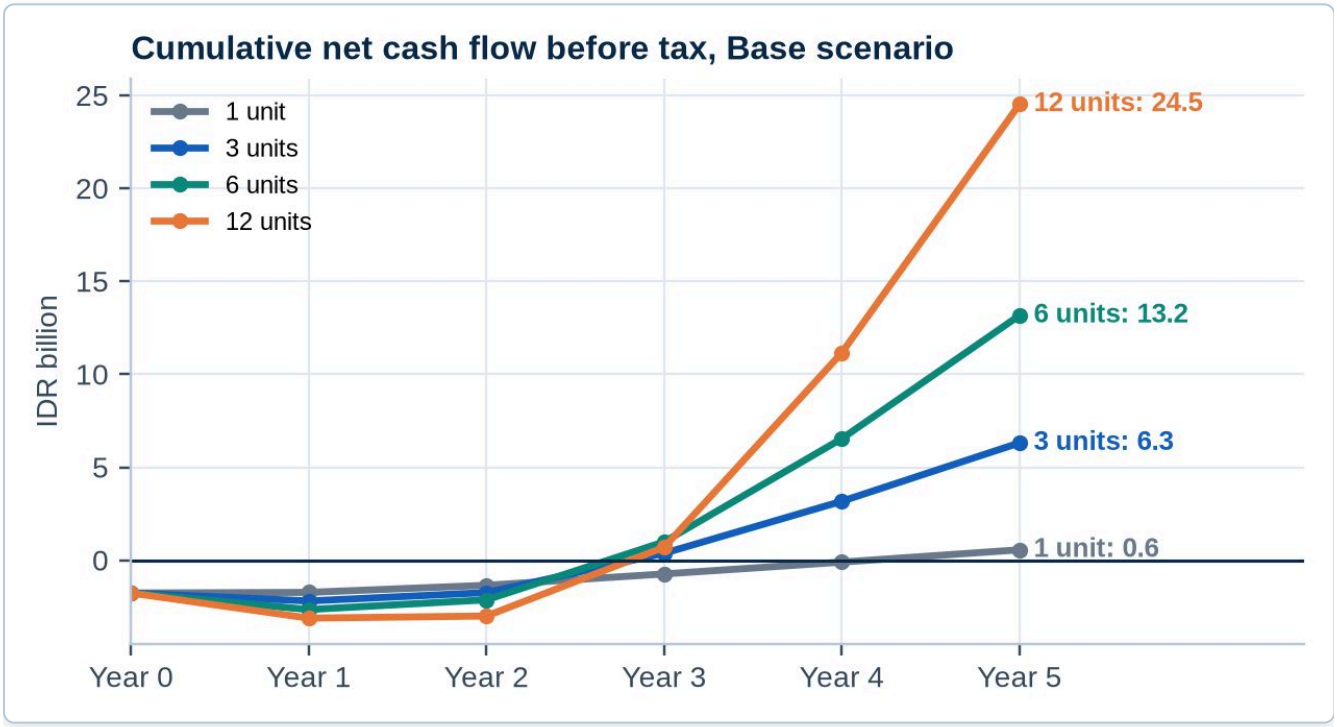



**Figure 7. Cumulative net cash flow before tax by programme size, Base scenario.**

Cash flow is negative through year 2 in every programme and recovers during year 3. A single unit reaches break-even only after **50 months**. When all drivers move together across the scenario switch, the three-unit NPV ranges from **-IDR 1,299 million** in the pessimistic case to **IDR 9,974 million** in the optimistic case. Section 6 shows how likely each outcome is.

# 6 Uncertainty and Risk

## 6.1 Sensitivity analysis

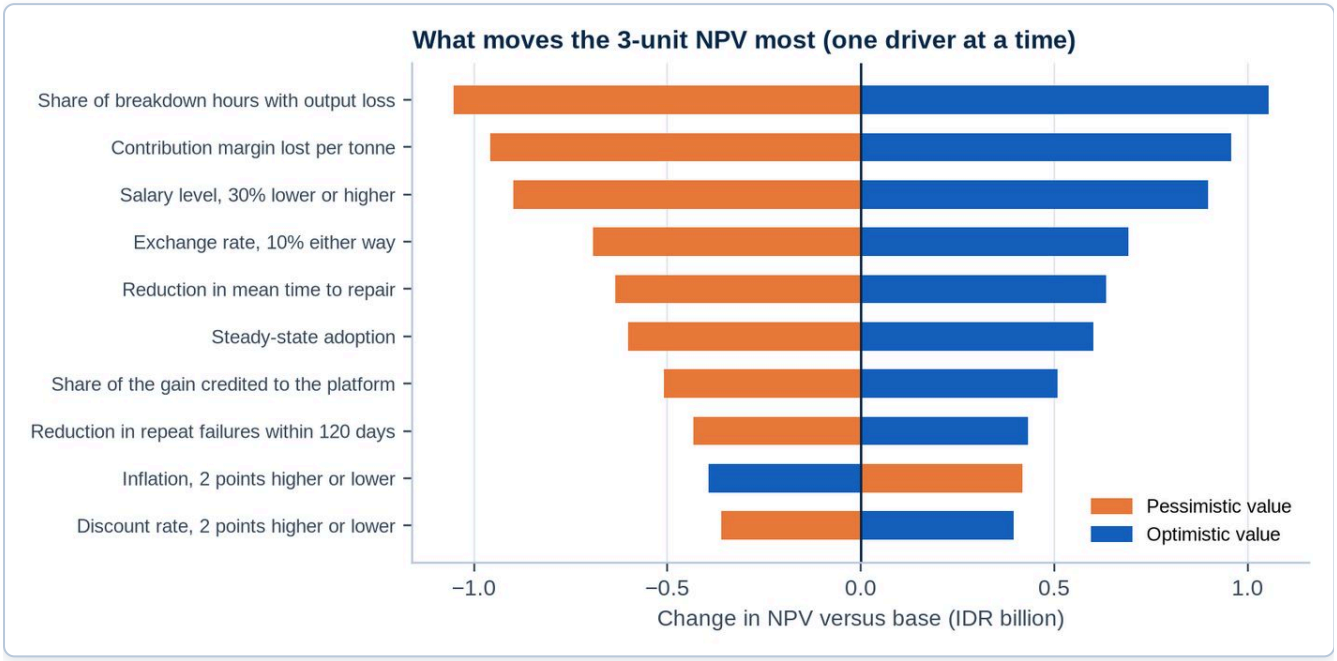



**Figure 8. One-at-a-time sensitivity of the three-unit NPV (Sensitivity sheet).**

The five largest drivers are the share of breakdown hours that reduce plant output, the contribution margin lost per tonne, the salary level (all roles), 30% lower or higher, the exchange rate, 10% stronger or weaker rupiah and the reduction in mean time to repair. The first two set the value of a downtime hour. The three-unit NPV remains positive until the share of breakdown hours that reduce output falls from 20% to **11%**, until the margin falls from USD 220 to **USD 116 per tonne**, and until the reduction in repair time falls from 10% to **2.2%**. Costs can rise by **48%** before the NPV reaches zero.

## 6.2 Monte Carlo simulation

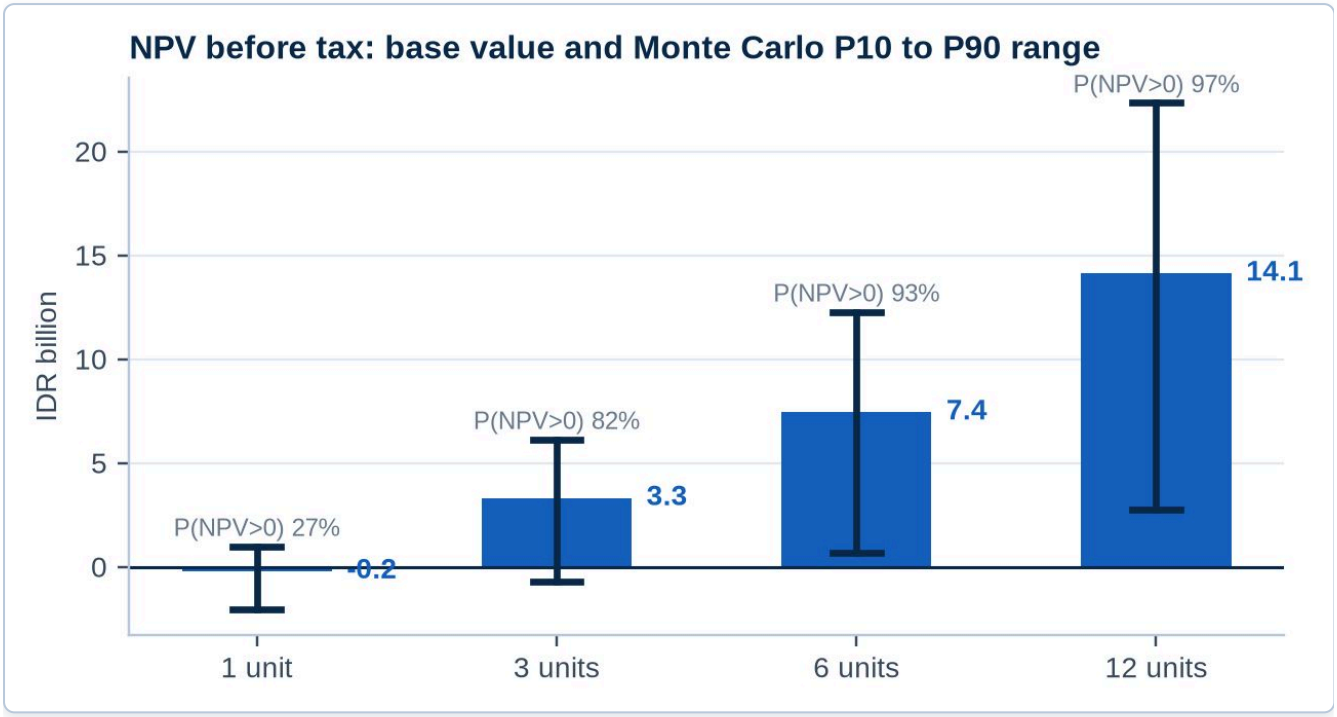



**Figure 9. NPV before tax with the Monte Carlo 10th to 90th percentile range (MonteCarlo sheet).**

The simulation draws 17 drivers independently from ranges that are wider than the scenario switch. For three units the mean NPV is **IDR 2,443 million**, the median is **IDR 2,066 million**, and **82%** of outcomes are positive. The 5th percentile is -IDR 1,418 million. A single unit is positive in only **27%** of outcomes, which confirms that the pilot should not be judged on its own return.

## 6.3 Risk register

**Risk register with financial impact (3 units, Base)**

*NPV impact is computed by shocking the Base scenario in the Python replica. Grey cells are static.*

| ID | Risk | Likelihood | NPV after shock (IDR M) | Change vs base (IDR M) | Shock applied in the model |
|---|---|---|---|---|---|
| R1 | Low adoption by operators and engineers | Medium | 255 | (3,065) | Steady-state adoption falls by 30% |
| R2 | Rollout delayed by one year | Medium | 1,753 | (1,567) | Units 2 and 3 go live one year later |
| R3 | One-off cost overrun | Medium | 2,551 | (769) | One-off cost 30% higher |
| R4 | Data quality is worse than expected | High | 1,825 | (1,494) | Ingestion effort doubles and benefits fall by 10% |
| R5 | AI service unavailable or withdrawn | Low | 2,501 | (819) | Search time reduction falls by 30% |
| R6 | Benefit attribution is challenged | Medium | 1,187 | (2,133) | Share credited to the platform falls by 30% |
| R7 | Polyethylene margin downturn | Medium | 766 | (2,554) | Contribution margin falls to USD 140 per tonne |
| R8 | Key person leaves the knowledge manager role | Medium | 1,354 | (1,966) | Adoption falls by 15% and running cost rises by 10% |
| R9 | Cybersecurity incident or audit finding | Low | 3,020 | (300) | Extra one-off cost of IDR 300 M |
| R10 | Data protection or contract constraints | Low | 2,917 | (403) | Connector cost 50% higher and security cost doubled |
| | **Combined stress: R1, R2, R3 and R6 together** | | (2,643) | (5,963) | |

**Figure 10. Risk register with the NPV effect of each shock (Risks sheet).**

The most damaging single risks are low adoption (NPV of IDR 255 million), a challenge to benefit attribution, and a margin downturn. A combination of low adoption, a one-year delay, a 30% cost overrun and weaker attribution gives an NPV of **-IDR 2,643 million**. The mitigations are role-based onboarding, measured baselines with control units, and the stage gates described in Section 7.

# 7 Benefits Realisation and Conclusions

## 7.1 Key performance indicators

| KPI | Baseline | Target | Measured by |
|---|---|---|---|
| Mean time to repair | 14.0 hours | 12.6 hours | CMMS work order dates |
| Jobs repeating within 120 days | 20.6% | 14.4% | Failure memory report |
| Minutes a day spent searching and checking | 40 | 20 | Pilot survey and timed tasks |
| Median time to a cited answer | About 15 minutes | 3 minutes or less | Timed pilot tasks |
| Answers with an approved source | Not measured | At least 90% | Answer audit log |
| Weekly active users | 0 | 85% of users | Usage log |
| Documents without approval status | 67% | 10% or less | Documents and Data Quality pages |

## 7.2 Stage gates

- **Gate 1, month 6.** Measured repair time, search time and adoption replace the assumptions. Proceed to unit 2 only if the weighted benefit reaches at least **70%** of the Base scenario.
- **Gate 2, month 12.** The repeat failure share and the answer audit are reviewed. Units 3 and beyond are approved against the sensitivity table.
- **Gate 3, month 24.** Full benefit review, model update and rollout plan.

## 7.3 Conclusions

- The investment case is positive from two units and strong at scale, with an NPV of **IDR 3.3 billion** at three units and **IDR 14.1 billion** at twelve units.
- The pilot is a learning investment. Its purpose is to replace assumptions with measured Company data before the programme grows.
- The result depends most on the value of a downtime hour. The first task of the pilot should be to measure how many breakdown hours reduce output at the real unit.
- The platform is priced for the value it creates inside the Company. External sales are an option once the pilot has proved the numbers.

**Decisions requested:** (1) approve the pilot budget of **IDR 1,766 million** subject to the three gates above; (2) nominate a knowledge manager and a unit sponsor; and (3) grant read access to EDMS, AIMS, CMMS and historian extracts for the pilot unit.

## 7.4 Limitations

The work orders and plant data are sample data prepared for the case. Salaries, margins, production loss and cost lines are estimates until the pilot replaces them. The model contains no terminal value, financing or working capital, and the Monte Carlo simulation treats drivers as independent. Market figures come from public summaries of analyst reports and should be read as orders of magnitude.

# 8 Appendix: Key Assumptions and Reconciliation

> Twelve inputs carry most of the uncertainty. Each is tagged by the quality of its evidence and is replaced by measured data during the pilot.

| Assumption | Base | Scenario range | Evidence tag |
|---|---|---|---|
| Minutes a day spent searching and checking | 40 min | 36 min to 44 min | Assumption |
| Reduction in search time | 50% | 45% to 55% | Assumption |
| Share of freed time that is productive | 60% | 55% to 65% | Assumption |
| Steady-state adoption | 85% | 80% to 90% | Assumption |
| Reduction in mean time to repair | 10.0% | 8.5% to 11.5% | Assumption |
| Share of the gain credited to the platform | 70% | 65% to 75% | Assumption |
| Reduction in repeat failures | 30% | 25% to 35% | Assumption |
| Contribution margin lost per tonne | USD 220 | USD 190 to USD 250 | Estimate |
| Share of breakdown hours with output loss | 20% | 17% to 23% | Assumption |
| Unit capacity (kt a year) | 336 | 336 to 336 | Market source |
| New joiners and role changes a year | 6 | 5 to 7 | Assumption |
| Platform configuration (person-months) | 8 | 9 to 7.5 | Estimate |

## 8.1 Reconciliation to the earlier slide

**Reconciliation to the earlier business impact slide**

*Earlier values are the figures shown on the previous slide (static, blue). New values are live links. Values at steady-state adoption.*

| Item | Earlier slide | New model | Change | What changed and why |
|---|---|---|---|---|
| Users per unit | 45 | 45 | 0% | Unchanged: 45 users in five role groups. |
| Search time lever, 1 unit (IDR M a year) | 604 | 340 | -44% | Loaded cost now built from role salaries, only 60% of freed time counts, and adoption is 85%. |
| Repair time lever, 1 unit (IDR M a year) | 857 | 528 | -38% | Downtime value is now built bottom-up. Only 70% of the gain is credited and repeat failures are removed first. |
| Repeat failure lever, 1 unit (IDR M a year) | - | 359 | new | New lever, based on the 120-day repeat rate in the work order data. |
| Ramp-up lever, 1 unit (IDR M a year) | 60 | 47 | -21% | Now built from joiners, days saved and loaded cost instead of a flat assumption. |
| Total value, 1 unit (IDR M a year) | 1,521 | 1,275 | -16% | Sum of the four headline levers at steady-state adoption. |
| Value of one breakdown hour (IDR M) | 30.0 | 33.1 | 10% | Capacity, margin, exchange rate and share of hours with output loss. |
| Share of freed search time that is productive | 100% | 60% | -40% | New haircut. |
| Share of repair gain credited to the platform | 100% | 70% | -30% | New haircut. |
| Pilot one-off cost (IDR M) | 1,400 | 1,766 | 26% | Cost anchor rebuilt line by line on the Costs sheet. |
| Pilot running cost (IDR M a year) | 370 | 731 | 98% | Cost anchor rebuilt line by line on the Costs sheet. |
| Payback, 1 unit (months) | 20.9 | 49.9 | 138% | Higher cost and ramp-up. A single unit does not repay quickly. |
| Value a year at 3 units (IDR M) | 4,562 | 3,825 | -16% | Year 0 prices at full rollout. |
| Payback at 3 units (months) | 11.8 | 33.9 | 186% | Phased rollout and adoption ramp are now modelled by cohort. |
| Net benefit over 3 years at 3 units (IDR M) | 7,930 | 375 | -95% | Earlier slide: net over three years. New: cumulative net cash flow to year 3. |
| Value a year at 6 units (IDR M) | 9,123 | 7,458 | -18% | Year 0 prices at full rollout. |
| NPV at 3 units (IDR M) | not shown | 3,320 | | New: discounted at the weighted average cost of capital. |
| IRR at 3 units | not shown | 44% | | New. |
| Repeat failures within 120 days | 8 of 55 | 13 of 63 | | Rule restated: same equipment and same failure mode in the previous 120 days, on all 63 corrective or breakdown jobs. |

**Figure 11. Reconciliation sheet. How each value on the earlier slide changed, and why.**

The workbook **Business_Impact_Financial_Model.xlsx** holds all inputs, calculations, cash flows, market data, sensitivities and references; the Sources sheet lists every reference with its link.